Frequently Asked Questions

Camera-related questions

1. Can two cameras (front and rear) be used simultaneously?

- SGG72 features dual integrated ISPs, enabling simultaneous operation of two RAW sensor camera modules.
- SOG42/52 features a single integrated ISP, supporting only one RAW sensor camera module. To enable simultaneous operation of two camera modules, either one or both must be smart sensor camera modules.

2. What camera configurations and resolutions are supported by the integrated dual ISP?

- Single camera 32MP @30 fps (dual ISP)
- Dual camera 16MP + 16MP @30 fps (dual ISP)
- Single camera 16MP @30 fps (single ISP)
- 3840 x 2160 @30 fps YUV422 8-bit video-in

3. Which camera sensor support on SOGx2 EVB

Item	OV5640	IMX219	IMX258	IMX708
Vendor	OmniVision	Sony	Sony	Sony
Effective Pixels	5MP	8MP	13MP	12MP
Maximum Resolution	2592×1944	3280×2464	4208×3120	4608×2592
Interface	DVP / MIPI	MIPI CSI-2	MIPI CSI-2	MIPI CSI-2
MIPI Lane	2 Lane	2 Lane	4 Lane	2 Lane
Internal ISP	Y	N	N	N

4. Raspberry Pi Camera Module support on SOGx2 EVB

No.	Part Number	Item Name	Description
1	SC0023	RPI CAM V2	CAMERA MODULE (IMX219 8MP)
2	SC1226	RPI CAM V3	CAMERA MODULE (IMX708 12MP WIDE NOIR)
3	SC1892	RPI5 CAMERA CABLE	CAMERA CABLE (200MM)

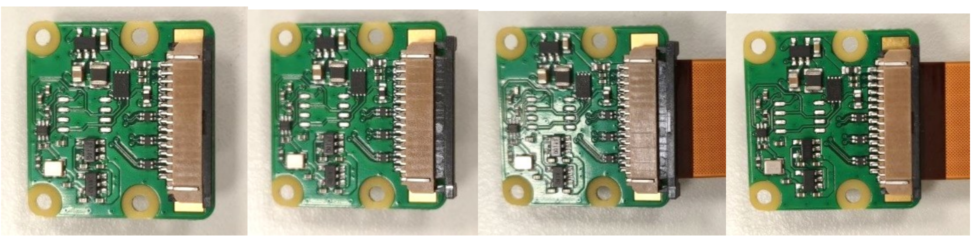
Camera Board Installation

Connect the V2/V3 Pi camera board to the CAM0 or CAM1 connector on the SOGx2 EVK using the camera FPC cable.

Installation procedure:

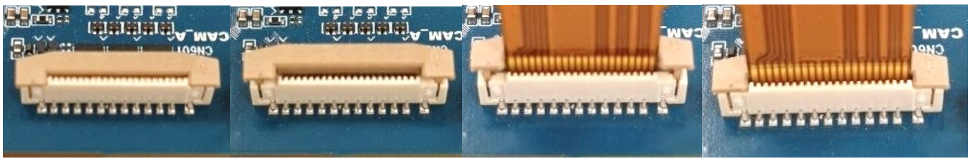
- **Step 1**

Loosen the camera board connector, insert the FPC into the camera board connector, and then lock it.



- **Step 2**

Loosen the connector on the SOGx2 EVK, insert the camera FPC into the connector, and then lock it.



Display-related questions

1. Video Decode and Encode Capabilities

- 3840 x 2160 @60 fps HEVC/H.264/VP9 video decoder (VDEC)
- 3840 x 2160 @30 fps HEVC/H.264 video encoder (VENC)

2. Which dual-display combinations are supported by the SOGx2 EVK? What is the maximum supported resolution for single-display?

The SOGx2 EVK supports eDP, DP, and DSI display interfaces. For single-display operation, a maximum resolution of 4K resolution is supported via either eDP or DP. The supported dual-display configurations are listed in the table below.

	Primary	Secondary
1	DSI(FHD60)	DP/DPoC(2.5K60)
2	DSI(FHD60)	eDP(2.5K60)
3	eDP(2.5K60)	DP/DPoC(2.5K60)
4	eDP(2.5K60)	DSI(FHD60)
5	DP(2.5K60)	DSI(FHD60)
6	eDP(4K)	
7	DP(4K)	

Currently, Android OS only supports Mirror Mode for dual display. Extended display mode is not supported.

AI development-related questions

What resources are available for AI development and performance evaluation on MediaTek Genio platforms?

Please refer to the following resources on MediaTek Genio platforms:

- **IoT AI Hub Overview**

  https://mediatek.gitlab.io/genio/doc/iot-aihub/master/ai_hub/overview.html

  The IoT AI Hub is the centralized entry point for evaluating, developing, and deploying AI workloads on MediaTek Genio platforms. It provides an end-to-end AI software stack, performance benchmarks, pretrained models, supported AI inference frameworks, and development tools to streamline the entire workflow—from model evaluation to edge AI deployment. It also includes benchmark results for validated AI models across different Genio SoCs, operating systems, and inference frameworks, enabling developers to assess AI performance and select the most suitable platform for their applications.

- **AI Benchmark for IoT Platforms**

  https://ai-benchmark.com/ranking_IoT.html

  For independent AI performance comparisons, the AI Benchmark IoT ranking provides standardized benchmark results for a wide range of embedded AI processors, including MediaTek Genio platforms. Developers can use these rankings to compare AI inference performance, CPU capabilities, and accelerator efficiency across different IoT SoCs, helping them evaluate the most appropriate hardware platform for their AI workloads.

Bootup-related questions

Auto power

By default, the SOG42/52/72 EVK is configured to power on automatically upon system power input. If manual power-on via a physical power button is required, please contact the sales representative to request this configuration.

Serial NOR Boot

Serial NOR Boot in Yocto provides a reliable way to store and execute bootloaders directly from NOR Flash, while leveraging eMMC/UFS for larger OS images. This hybrid approach balances fast, stable booting with high-capacity storage. For more details, refer to the following documentation:

https://genio.mediatek.com/doc/iot-yocto/latest/sw/yocto/poc/norboot.html